ABES Engineering College, Ghaziabad
(An Autonomous College)
Department of Applied Sciences &Humanities

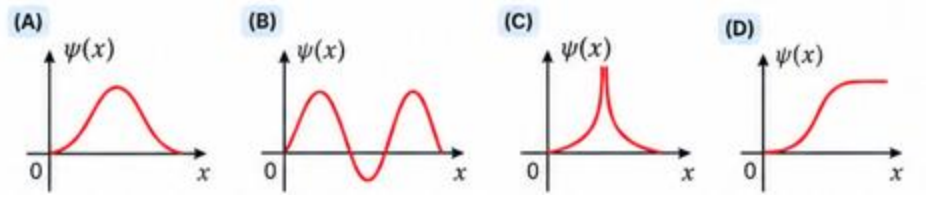
Program Name: B.Tech **Session: 2026-27**

Semester: I **Section: CSE1-CSE7, AIML1-AIML6, ELCE1-ELCE2** **Max. Marks: 10**

Subject Code: 25AS101 **Subject Name: Applied Physics**

Date of Assignment: 25/9/2026 **Date of submission: 1/10/2026** **SL & TW Hours: 5**

ASSIGNMENT NO.1

Q. No.	Question	CO	KL
1	State reasons for need of quantum mechanics?	CO1	K1
2	Derive the expression for the de Broglie wavelength of a particle in terms of its linear momentum.	CO1	K3
3	What is the difference between matter waves and electromagnetic waves?	CO1	K1
4	Define wave packet. Find the relation between phase velocity and group velocity.	CO1	K3
5	Derive the Eigen value and eigen function for a particle in one dimensional potential box of width L.	CO1	K3
6	Which of the following wave function is not physically acceptable and why? 	CO1	K1
7	Which of the following functions cannot represent a physically acceptable wave function for all values of x? (a) Ψ= A sinx (b) Ψ= A cosx (c) Ψ= A tanx (d) Ψ= A e^{-x^2}	CO1	K1
8	Signify the wave function. Also deduce the Schrodinger equation for free particle.	CO1	K3
9	What are the fundamental principles of quantum computing? Define quantum bit in quantum computing.	CO1	K1



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10	Write a short note on Qunatum cryptography.	CO1	K1
11	A particle is moving in 1-D box of width 25Å. Calculate the probability of finding the particle within an interval of 5Å at the center of the box when it is in its state of least energy.	CO1	K3

KL- Bloom's Knowledge Level (K1, K2, K3, K4, K5, K6)
K1 – Remember K2 – Understand K3 – Apply K4 – Analyze K5 – Evaluate K6 – Create